

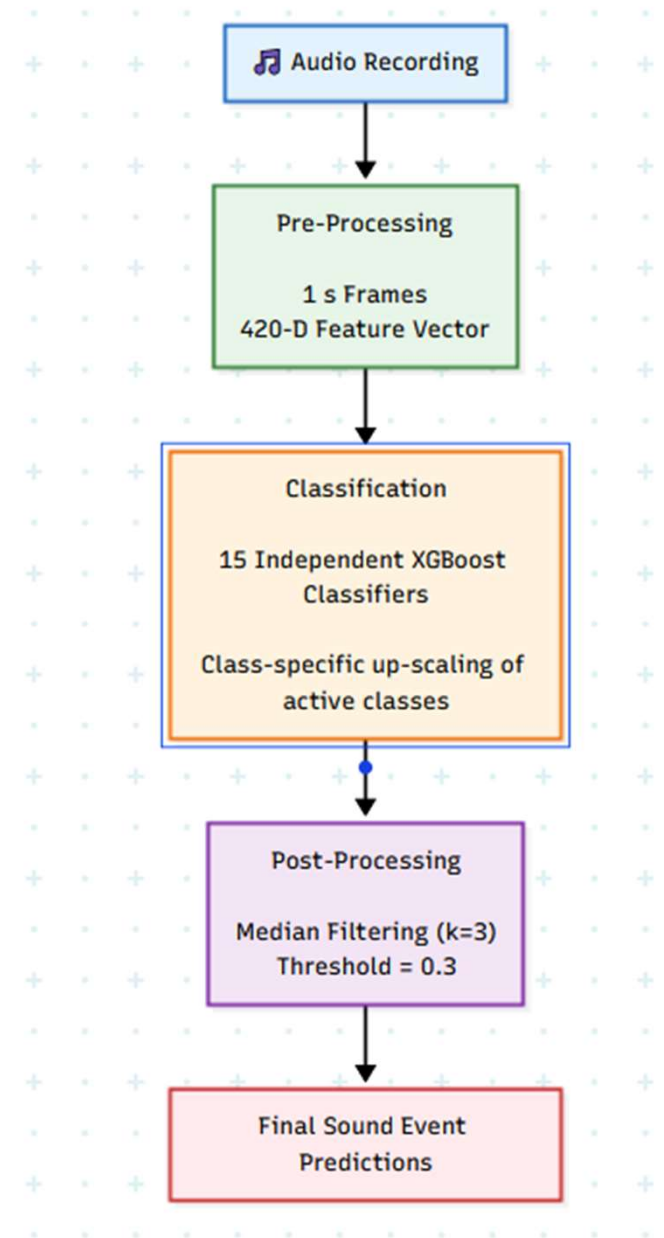
Machine Learning Pattern Classification - Challenge

Temporal Frameworks

Sebastian Pichler & Laurin Siebert

Model Architecture

- Trained on 420 Feature Dimensions
- Multi-label setup via independent binary classifiers
- Class specific positive weighting
- Temporal smoothing as post-processing via Median Filtering with a (3,1)-sized Kernel



Investigated Hypotheses

- Event duration strongly affects detection performance
 - Long stationary sounds consistently achieve highest performance
 - Short transient sounds are averaged out in 1s windows and a median filtering with a larger kernel
- Acoustic distinctiveness matters more than frequency alone
 - Although light switch is a rare and short sound event using this model structure it became well detectable
- Median filtering improves robustness of the prediction
 - A larger Kernel size correlates with higher recall
 - The sweetspot does not lose precision, but still gains robustness
 - Removes isolated noise and fills gaps for sustained events

Conclusion Summary

- XGBoost significantly outperforms the baseline
- Class (specific) weighting helps rare events
- Acoustic distinctiveness is more important than raw frequency
- Temporal smoothing removes prediction noise

Limitations

- Too large hop size / temporal resolution for short events
- Confusion between similar events. E.g. door_, wardrobe_drawer_ and window_open_close
- Independent classifiers ignore inter class correlations

Future Directions

- Shorter frame hop size
- Learn audio embeddings to differentiate nonlinear clusters
- CRNN / Transformer based SED models
- Multi-label architecture to detect co-occurrence